EL NIÑO ACTION BRIEFING · 28 SEPTEMBER 2026 · VERIFIED RUN

Uvita, Osa, Costa Rica

The risk here this winter is the tap, not the thermometer. A spring-fed water system meets its first strong El Niño at the height of tourist season; lock in the water plan and stop adding demand before November.

IMPACT REGION	EXCESS HEAT DEATHS	WINDOW	DOMINANT HAZARD
Not resolved — Osa canton (~29,500) sits inside a larger, unconfirmed region	No figure published for Costa Rica or Central America	n/a (study's Sep 2026 – Feb 2027 window has no local figure)	Drought and water supply; secondary: heat, reef heat stress, late-season rain

16/16 recommendations checked. 0 confirmed without correction, 10 corrected, 3 unverified, 3 contraindicated. Exposure figure: Climate Impact Lab, *451,000 at Risk*, September 2026, pre-peer-review — it publishes no figure for Costa Rica, and nothing in this report is scaled from it. The historical record and IMN/CNE bulletins carry the hazard instead. Three corrections to the starting brief: Uvita is a town in **Bahía Ballena district** (≈3,300 people, INEC 2024 estimate), not a district of its own; the Pacífico Sur rainfall deficit is **25%** (Jan–Aug), not 35%; and Osa is on CNE **yellow** alert, with **no 2026 drought decree** covering it. The Climate Impact Lab Adaptation Inventory could not be consulted (its download sits behind a sign-up form), so no action below is marked as having or lacking a precedent there.

EXECUTIVE SUMMARY — AT A GLANCE



9.16°N 83.74°W, Costa Rica · Natural Earth

- **The risk.** The tap, not heat: a spring-fed system that rationed by the hour around 2012 meets its first strong El Niño with no margin left; Osa's recorded deaths came from non-El-Niño storms.
- **First action.** ASADA Uvita–Bahía, with AyA's regional ASADA office and the Osa emergency committee, publishes a dry-season water plan by 15 November 2026.
- **The money.** $40k floor, $106k target, $136k ceiling. Top placement: $40k to ASADA Uvita–Bahía for leak repair, chlorination, and backup power, released against an itemised quote.
- **Do not.** Do not tell older people fans are safe up to about 40 °C: the cited Lancet review advises against fans above 35 °C for people 65 and over.*

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* Suggested formulation: check with a health worker before sharing.

WHAT THIS PLACE SHOULD DO — RANKED

Ranked by expected lives saved per unit of effort before the peak. Backing lenses are shown per action; an action several lenses reached independently is the stronger bet. Deaths here are rare enough that no action's effect will be measurable in deaths — track outages, dengue admissions, and workers' kidney markers instead. The RACI chart below assigns each one.

1. Publish a dry-season water plan for ASADA Uvita–Bahía by 15 November

PARTIALLY CONFIRMED · EMERGENCY · HISTORIAN · SKEPTIC · FUNDER · ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
ASADA Uvita–Bahía, with AyA's regional ASADA office and the Osa emergency committee	15 Nov 2026	Within operating budgets (no published figure)	Unplanned outages at peak demand; rationing that falls on the poorest first

Measure spring flows weekly, repair leaks, publish a staged rationing schedule that guarantees a minimum volume per household, line up tanker supply, and brief hotels before December. The ASADA serves ~1,760 connections and has rationed by the hour before (~2012) when building outran supply; its 2017 expansion was expected to cover about five years of growth. AyA's national plan starts controlled rationing once source deficits pass 17%, and 6,525 services in 74 communities elsewhere are already restricted.

Correction: the 17% is AyA's national threshold, not an ASADA playbook. No study shows pre-agreed rationing prevents inequity; the logic is sound but the effect is unmeasured.

2. Stop adding demand: suspend new water-availability letters on a technical supply study, and ask MINAE to restrict new wells

PARTIALLY CONFIRMED · SKEPTIC · ECONOMIST · ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
ASADA board; MINAE Dirección de Agua	1 Nov 2026	Lost connection-fee revenue (unquantified)	New hotel and villa connections drawing on springs as they fall

* Suggested formulation: check with a health worker before sharing.

Costa Rica's Constitutional Court has upheld availability-letter denials backed by a technical study, and in ruling 2018-002898 stopped an ASADA and AyA from issuing letters over an overexploited aquifer in Potrero, Guanacaste. Well permits sit with MINAE's Dirección de Agua (DE-43053-MINAE), with precedent for coastal restrictions in Carrillo. The Skeptic's test: whether the ASADA will refuse new tourist connections in November. Nobody has asked it to.

Correction: a new dry-season surcharge on heavy users cannot be approved before the peak — ARESEP sets ASADA tariffs through a public hearing (the last general adjustment took three months, Nov 2024 to Feb 2025). Use the existing schedule this season; file an ASADA-specific tariff petition for next year.

3. Install ~300 sealed, screened household tanks as outage buffers before December

PARTIALLY CONFIRMED — PURPOSE CHANGED | HISTORIAN · FUNDER · FRONTLINE; SKEPTIC DISSENT RESOLVED | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
ADI de Uvita, households selected through IMAS or the ASADA's outage-risk map	**30 Nov 2026**	**~$66k–90k (300 × $220–300 delivered and fitted)**	**Households going dry during rationing or pipe outages**

Retail price for a 1,100 L tank is ₡97,450–105,000 (≈$195–210); fittings, screening, and delivery to Osa bring it to roughly $220–300. A 1,000 L tank holds two to four days of household use. Tight or net lids cut Aedes larvae and pupae in household containers in field trials in Central America and Vietnam.

* Suggested formulation: check with a health worker before sharing.

Correction: these are *network* buffers, not rain supply. Uvita averages under 50 mm a month from January to March, so a tank filled in October is empty long before April. The 2019 Bahía Ballena rainwater pilot was three sites, and its $358 figure was for one 450 L system, not a 1,000 L tank. SINIRUBE, the registry the funder proposed for targeting, is probably restricted to public institutions — route selection through IMAS or the municipality.

4. Time a container-control push with the Ministry of Health's vector programme, and keep it running into June 2027

PARTIALLY CONFIRMED | PUBLIC HEALTH · SKEPTIC | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
EBAIS / CCSS health area and Ministerio de Salud, with the ASADA	Start before any rationing; run to Jun 2027	Within the national programme	A dengue outbreak when the rains return

The Camino Verde cluster RCT (Andersson et al., BMJ 2015, Nicaragua and Mexico) cut dengue infection in children by 29.5% (95% CI 3.8–55.3) and houses with larvae or pupae by 44.1% through community mobilisation to cover and empty containers; the larvicide temephos added nothing. The danger is the lag: the dry season should lower dengue, but containers filled for rationing are still standing when the rains return around May.

Transferability is good (Central American households). Correction: the Ministry already runs a national programme (1.1 million breeding sites eliminated by week 38 of 2025), so this is coordination and timing, not a new programme. The drought-to-dengue lag comes from a Brazilian urban modelling study, weaker for a town of 3,300. No Osa dengue data was retrievable.

5. Keep chlorination running through any intermittent supply; fund a generator only if pumps are confirmed

UNVERIFIED AS SCOPED — RECAST | SKEPTIC · PUBLIC HEALTH · FUNDER | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
ASADA Uvita–Bahía	Itemised quote by 31 Oct	Up to $40k, only against an itemised quote	Contamination when pipes lose pressure

* Suggested formulation: check with a health worker before sharing.

Ercumen et al. (PLoS Medicine 2015, Hubli-Dharwad, India) found E. coli in under 1% of tap samples under continuous supply versus 32% under intermittent supply, and lower typhoid (CIR 0.58), though overall diarrhoea did not differ significantly. Check chlorine residual daily during rationing, flush lines when supply resumes, and give households guidance on safe storage.*

The system is springs plus a 500 m³ tank, but it also has a pumped line under the Río Uvita (built 2019, paid by Inder and AyA), so pump dependence is plausible and unconfirmed. The ASADA holds AyA's four-star potability seal, so the baseline risk is low; the Hubli networks were large, old urban systems.

6. Enforce the existing heat-stress decree with palm, construction, and farm employers before December

PARTIALLY CONFIRMED — RECAST AS ENFORCEMENT | PUBLIC HEALTH · FRONTLINE | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Ministerio de Salud and MTSS inspectors; employers	1 Dec 2026	Inspector time; water, shade, and rest are an employer cost	Acute kidney injury and heat illness in outdoor workers

Decreto 39147-S-TSS (2015) already requires a hydration, shade, rest, and protection protocol for outdoor workers. In Nicaraguan sugarcane (Hansson et al., OEM 2025), a water-rest-shade programme cut acute kidney injury from 20 to 8 per 1,000 worker-months, and incident kidney injury from 21% to 1%. Carry the message through foremen and palm buses, in Spanish, with no questions about immigration status.

The evidence is a non-randomised cohort of burned-cane cutters (n=1,044), a harsher job than palm or construction. The often-quoted $1.60 return per $1 is one year's figure; the five-year average was $1.02.

7. Keep road-cut readiness live through the end of the rains, then stand it down

PARTIALLY CONFIRMED | EMERGENCY · HISTORIAN | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Osa municipal emergency committee (CME)	Stock placed by 31 Oct; flood roster live through Nov	Redistribution of existing stock	A repeat of 2017, when Ruta 34 and the Interamericana both went down

* Suggested formulation: check with a health worker before sharing.

Place medical, water, food, and shelter stock between the likely cut points on Ruta 34 and the Interamericana, and plan as if Uvita is cut off for 72 hours. In 2017 Tropical Storm Nate isolated Hospital Tomás Casas and flooded Ciudad Cortés; in 1996 Hurricane César left 10 missing in Osa.*

Correction: none of those disasters happened in an El Niño year (NOAA ONI: 1996 neutral, 2008 La Niña, 2017 neutral turning La Niña; 2026 is already strong El Niño). El Niño dries the Pacific side, so hold the posture through November's rains, not through the drought. The CME's page names no member agencies, and a Cruz Roja station near Uvita was not verified.

8. Reach the households the ASADA's WhatsApp line never will

PARTIALLY CONFIRMED · FRONTLINE · ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Volunteers through ADI de Uvita and churches	Saturdays in Oct and Nov	Volunteer time and printing	Off-network households going unwarned and unserved

Relay the ASADA's notices into Spanish WhatsApp groups and English expat Facebook groups, and use a market table to map who is not on the network: hill homes on private springs, migrant workers in shared rentals, and English-only retirees. Hand out a one-page sheet in both languages covering safe storage, heat-illness signs, and where the EBAIS is. Feed the map into action 3.

The ASADA's channels are confirmed (WhatsApp +506 8736 7004, tel 2743 8050), though it has no dedicated outage channel. The Saturday market and any water operators for Ojochal, Dominicalito, and the hill communities were not verified.

9. Get the money gates open early: help households self-register for IMAS, and ask CNE to recommend Osa for any drought decree

PARTIALLY CONFIRMED · ECONOMIST · FUNDER · ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
ADI de Uvita and volunteers (IMAS registration); Osa municipality (CNE request)	31 Oct 2026	Staff and volunteer time	Emergency money arriving after the harm

* Suggested formulation: check with a health worker before sharing.

IMAS pays emergency transfers on an incident report from CNE, the municipal committee, Bomberos, Cruz Roja, CCSS, or the Ministry of Health — no decree needed — but only to households already registered in SINIRUBE, and households must register themselves. For larger money, the President declares emergencies by decree on CNE's recommendation; the Fondo Nacional de Emergencias and the World Bank's US$160M Cat DDO (which does cover drought) both pay only after that. The 2019 drought decree added Coto Brus from this region, which is the precedent.

No legal deficit threshold triggers a decree, so any threshold is an agreement, not a rule. IMAS does not name drought or give an amount; the Cat DDO's drawdown window may need renewal. Article numbers in Ley 8488 were not verified.

10. Build a volunteer check-in list for isolated older residents, and correct the fan advice

PARTIALLY CONFIRMED — LOW TRANSFERABILITY | PUBLIC HEALTH · FRONTLINE · FUNDER | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
ADI de Uvita, churches, and expat groups (opt-in list)	1 Dec 2026, check-ins to April	Up to $10k (held in the ceiling)	A handful of heat or outage deaths at most

Call isolated older Ticos in the hills and older expats living alone every two to three days during outages or heat peaks. Fans help below about 35 °C; above 35 °C, people 65 and over should use skin wetting or a cooler space instead (Meade et al., Lancet Planetary Health 2024).*

* Suggested formulation: check with a health worker before sharing.

Ahmedabad's heat action plan (Hess et al. 2018) is associated with about 1,190 fewer deaths a year, but it was a whole-city plan for ~7 million people at 47 °C. Nothing in that transfers to a town of 3,300 with maxima near 33 °C. An EBAIS patient roster probably cannot be shared under data-protection law, so build an opt-in list instead.

11. Take the non-heat stress off Marino Ballena's reef while it bleaches

PARTIALLY CONFIRMED — LIVELIHOOD, NOT LIFE-SAVING | HISTORIAN | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Osa municipality and SETENA (erosion control); SINAC (visitor limits)	Now, while it is still raining	Enforcement time	Bleaching turning into coral death

NOAA Coral Reef Watch reads heat stress on Costa Rica's Pacific coast at about 15.9 °C-weeks and rising (27 Sep 2026), with sea temperature 30.6 °C against a warmest-month normal of 28.8 °C. In 1991–92, 57% of the park's colonies bleached but only 9% died when other stresses were low. Enforce erosion control on hillside building sites and limit visits to stressed sites.

Sediment is the most likely cause of the dead 0.87 ha reef near Tres Hermanas, but that is a hypothesis, and Quebrada Mercedes is named as the source at Playa Ballena only (Alvarado et al. 2005). This protects the tourism economy, not lives.

12. Restrict agricultural burning and stand up a fire brigade for the dry season

UNVERIFIED | EMERGENCY · FUNDER | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Not confirmed (burn-permit issuer unverified)	Dec 2026 – Apr 2027	Unfunded	Grass and forest fires on the Fila Costeña

The direction is sound: El Niño dries the Pacific side and the dry season is fire season. But no source confirmed who issues burn permits here, a volunteer brigade in Osa, or recent fire history for the canton.

* Suggested formulation: check with a health worker before sharing.

SINAC's national fire coordination covers the nine conservation areas with the most fires, and nothing shows Osa is one of them. The funder's $10k brigade kit is cut to $0 until SINAC's Osa conservation area (ACOSA) confirms in writing that a brigade exists.

* Suggested formulation: check with a health worker before sharing.

R = Responsible (does the work) · **A** = Accountable (one per action; signs off and answers for it) · **C** = Consulted (asked before) · **I** = Informed (told after). Built from the owners named in the verified actions. No one here has been contacted or has agreed to a role; every row is a proposal.

Action	ASADA Uvita–Bahía	AyA	MINAE Dirección de Agua	ADI de Uvita	IMAS	EBAIS / CCSS	Ministerio de Salud	MTSS inspectors	Employers	Osa emergency committee (CME)	Osa municipality & SETENA	CNE	Volunteers, churches & expat groups	SINAC (ACOSA)
1. Dry-season water plan by 15 November	**A/R**	**C**								**C**				
2. Suspend availability letters; restrict new wells	**A/R**	**C**	**R**											
3. ~300 sealed household outage-buffer tanks	**C**			**A/R**	**C**								**C**	
4. Container-control push into June 2027	**R**					**R**	**A**							
5. Chlorination through intermittent supply	**A/R**													
6. Enforce the heat-stress decree							**A**	**R**	**R**					
7. Road-cut readiness through the rains						**I**				**A/R**		**I**		
8. Reach households off the WhatsApp line	**C**			**A**		**I**							**R**	
9. IMAS self-registration; ask CNE about Osa				**A/R**	**C**						**R**	**C**	**R**	
10. Check-in list for isolated older residents				**A/R**									**R**	
11. Cut non-heat stress on the reef											**A/R**			**R**
12. Restrict burning; dry-season fire brigade														**A/R**

What the chart shows. **The ASADA is accountable for all three water-supply rows and is consulted on two more**, so the plan depends on one volunteer-board utility with no published budget for any of it. **ADI de Uvita is accountable for four rows**, and on three of them the work falls to volunteers, churches, and expat groups whose only money is $10k held in the ceiling. **Row 9 is split**: IMAS registration sits with ADI de Uvita, while the request to CNE sits with the Osa municipality, so the chart puts the A on ADI de Uvita and an R on the municipality. **Row 12 has no named owner in the report**: the burn-permit issuer is unverified, so the A sits on SINAC's Osa conservation area, which the text says must confirm a brigade exists before any money moves.

* Suggested formulation: check with a health worker before sharing.

The funder lens, reconciled against the ranked actions and verified. Placements are ranked by marginal lives saved per dollar. The envelope is bounded by what this place can spend before the peak, not by what it needs.

FLOOR	TARGET	CEILING	LARGEST UNCERTAINTY
$40k — the ASADA placement, against an itemised quote	**$106k — ASADA $40k + tanks $66k**	**$136k — target + check-ins $10k + cash reserve $20k**	**Whether the springs actually fail; the only flow record found is from 2015**

Arithmetic: floor = $40k; target = $40k + (300 × ~$220) = $106k; ceiling = $106k + $10k + $20k = $136k. The target moved from $134k to $106k in verification (−21%, under a third): the tank cost rose from $54k to $66k, the $10k fire kit was cut to $0, and the check-ins and cash reserve moved to the ceiling because no delivery organisation was confirmed. The funder's $175k ceiling rested on its own capacity estimates and is not used. If tanks are fitted at the upper price ($300), the target is $130k.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
ASADA Uvita–Bahía	Actions 1 and 5: leak repair, chlorination, backup power if pumps are confirmed	$40k	Direct grant, released against an itemised quote	Equipment estimate. The cited $28–72 per DALY (Kremer et al. 2022, NBER) comes from high child-mortality settings and does not transfer to a four-star-certified system, so it is dropped.	PARTIALLY CONFIRMED
ADI de Uvita	Action 3: ~300 sealed outage-buffer tanks	$66k	Pre-positioned stock	300 × ~$220 delivered (1,100 L retail ₡97,450–105,000). A cost per household reached, not per death averted.	PARTIALLY CONFIRMED
ADI de Uvita or Fundación SOMOS	Action 10: check-in list	$10k (ceiling only)	Matching grant	No local evidence of heat deaths; roster access blocked.	UNVERIFIED
Not named	Cash to households if the ASADA starts rationing	$20k (ceiling only)	Triggered unconditional cash	Sensible trigger, but no delivery organisation, and it may duplicate IMAS emergency transfers (action 9).	UNVERIFIED

* Suggested formulation: check with a health worker before sharing.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
SINAC-ACOSA	Action 12: fire brigade kit	$0 (was $10k)	—	No brigade confirmed in Osa.	UNVERIFIED

The funder's point, which survived verification: on lives saved per dollar, Uvita is a weak placement. Residents have universal CCSS coverage, a four-star certified water system, no rationing yet, and no decree. Keep money here small and trigger-based; the larger money belongs with the northern Pacific ASADAs that are already rationing. The proposed condition that Envision Festival match any grant was dropped: Envision has made no dollar commitment, only a pledge to build its own water system with the ASADA so its March festival draws less from the town.

DO NOT DO THIS — CONTRAINDICATED

Advice that circulates for this hazard and would do harm or waste the window here.

"Fans are safe for older people up to about 40 °C"

CONTRAINDICATED

The cited review (Meade, Notley, Kirby & Kenny, Lancet Planetary Health 2024) says the opposite: agencies should keep advising against fans above 35 °C, especially for people 65 and over, and it found no reliable core-temperature benefit above that. Uvita's usual 30–35 °C is mostly below the line, which is exactly why the exception gets forgotten on El Niño's hottest days.

Instead: fans below 35 °C; above it, skin wetting and a cooler space for older and chronically ill residents.*

Rain tanks or open barrels as the answer to dry-season supply

CONTRAINDICATED

The El Niño peak falls in a dry season averaging under 50 mm a month from January to March, so rain tanks stand empty when they are needed. Unscreened tanks are documented Aedes breeding sites (Beebe et al., PLoS NTD 2009), and open storage during drought sets up the dengue lag the epidemiologist warns about.

* Suggested formulation: check with a health worker before sharing.

Instead: sealed, screened tanks filled from the network as outage buffers (action 3), with container control through the return of the rains (action 4).

Waiting on an IFRC drought grant or a parametric insurance payout

CONTRAINDICATED

No IFRC DREF or early-action protocol for drought exists for Costa Rica. The one active protocol (EAP2023CR02) covers tropical-cyclone floods only, triggered by an IMN forecast above 400 mm in five days. An imminent-crisis DREF for a yellow-alert canton is implausible. Costa Rica is not a CCRIF member, no parametric wage or heat cover exists there, and a fishers' product keyed to sea temperature would need design, SUGESE approval, and reinsurance — none of it possible before April. The uninsured loss here is water, not wages.

Instead: the gates that exist — IMAS self-registration now and a CNE recommendation for Osa (action 9) — and local money for the ASADA.

Drilling private or hotel wells as drought insurance

ADVISE AGAINST — CONTRAINDICATION NOT VERIFIED AT PRIMARY SOURCE

The ASADA left its own wells after houses built around them threatened water quality, and new wells on this coast risk saltwater intrusion and draw down the supply everyone shares. Guanacaste's coastal aquifers show the pattern. A primary SENARA or Dirección de Agua source was not retrieved this run, so this stays a strong caution rather than a verified finding.

Instead: a local restriction on new wells requested from the Dirección de Agua (action 2).

WHAT HAS HAPPENED HERE BEFORE — THE RECORD

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
1991–92	El Niño coral bleaching, Marino Ballena	57% of colonies bleached, 9% died	Reef cover kept falling through later El Niños

* Suggested formulation: check with a health worker before sharing.

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
1996 (Jul)	Hurricane César — ENSO neutral	11 dead and ~50 missing nationally; 10 missing in Osa; Interamericana Sur cut	Nothing specific found
2008	Tropical Storm Alma — La Niña	2 dead in Costa Rica; ~$35M damage; 100+ roads and bridges	Nothing specific found
~2012	ASADA Uvita–Bahía rations by the hour as building outpaces supply	No health toll recorded	Wells abandoned; six springs added 2012–15; 500 m³ tank in 2017, sized for ~5 years
2014	National drought emergency decree	—	Osa not among the five Puntarenas cantons included
2017 (Oct)	Tropical Storm Nate — ENSO neutral	National deaths reported 8–28; Hospital Tomás Casas isolated; Ciudad Cortés flooded	Nothing specific found
2019	Drought decree	—	Coto Brus added from Pacífico Sur; Osa not
2023–24	El Niño drought	32 ASADAs rationing in Guanacaste, Puntarenas, and Alajuela; none documented in Osa	—
2026	Strong El Niño; Pacífico Sur −25% rainfall	Osa on yellow alert; no decree; reef heat stress ~15.9 °C-weeks	This briefing

The recurrence nobody has noticed: in Uvita the reef and the tap fail for the same reason, and El Niño only exposes it. Development since the Costanera opened puts sediment on the reef and houses over the wells. The water system's last fix was sized for about five years from 2017, so this is likely the first strong El Niño to reach it with no margin left. Drought is the trigger; growth makes it a shortage. No drought death has ever been recorded in Osa — the deaths here have come from storms in non-El-Niño years.

* Suggested formulation: check with a health worker before sharing.

WHERE THE EXPERTS DISAGREE — UNRESOLVED

- **Plan as an island through the drought** versus **El Niño makes the south Pacific dry, so no Nate-scale flood is coming**. The emergency manager holds the first, the historian the second. NOAA's ONI settles most of it: none of the three deadly storms came in an El Niño year. The residual risk is short, concentrated rain on steep ground, so keep the posture through November and stand it down for the dry season.
- **Fund tanks** versus **tanks breed dengue and stay empty**. The historian, funder, and outreach lens want tanks; the skeptic warns against them. Both are right about different tanks: sealed network buffers yes, rain supply and open barrels no.
- **Put money in Uvita** versus **send it to the rationing north**. Every local lens designs for Uvita; the funder says Uvita is a weak lives-per-dollar placement. What would settle it: a weekly spring-flow series from the ASADA. If flows are holding in November, the funder is right.
- **Dry season lowers dengue** versus **water storage raises it**. The epidemiologist and the skeptic agree on the mechanism and differ on timing: the risk peaks after the drought, when the rains return around May.

WHAT NOBODY COVERED — BLIND SPOT

Visitors. The El Niño peak is Uvita's high season, and Envision Festival brings thousands of people to the same water system in March, the driest month. No lens looked at heat illness, water demand, or evacuation for a mass gathering in drought, and none checked electricity: Costa Rica's grid is largely hydro, and a dry year that forces power cuts would also stop any pumped part of the water network. Either could change the ranking — a festival-week water plan may matter more than any household measure.

HOW TO READ THIS — METHOD

The panel is author-built. Seven simulated expert lenses were used to generate these recommendations. Where they agree, that is a strong hypothesis, not a consensus of the field. Every recommendation was then independently checked for evidence, transferability to this location, and feasibility before the peak; the verdict tags carry the result. Several checks ran out of web-search budget partway through, which is why some items remain unverified rather than confirmed.

Mortality figures are resolved at Climate Impact Lab impact-region scale, roughly 300,000 people. They are not neighbourhood figures. The El Niño projection covers *heat only*; drought, flood, fire, smoke, and crop failure are additional to it and are carried here by the historical record rather than by a published number. For Costa Rica no figure is published at all.

* Suggested formulation: check with a health worker before sharing.

- **Scope.** The location was taken as given: Uvita, Osa, Costa Rica. Climate Impact Lab impact region, as the report records it: Not resolved — Osa canton (~29,500) sits inside a larger, unconfirmed region. Mortality figures exist only at that scale (impact regions average roughly 300,000 people); sub-city detail is used for hazard exposure and action design, not for death counts.
- **Exposure.** Climate Impact Lab, *451,000 at Risk* (September 2026, pre-peer-review): ECMWF SEAS5 seasonal forecasts (51-member ensemble) applied to temperature–mortality relationships across 24,378 regions, against a 1996–2025 baseline, for June 2026 – February 2027. It covers heat only. It is never added to the Impact Map's long-run projections or quoted from the map's GDP-denominated mortality. Where heat is not the dominant hazard, the historical record carries the hazard instead of a borrowed number.
- **Expert panel.** Seven simulated expert lenses (emergency manager, public health scientist, frontline responder, skeptic, economist, historian, and philanthropist) each researched this place independently and in parallel, from web sources, to a fixed brief.
- **Reconciliation.** Duplicate actions were merged and backing lenses recorded. Convergence was tested for shared bias (whether the evidence behind an agreed action came from a setting like this one) before it counted in an action's favour. Contradictions were kept, not resolved by dropping a side. Actions were ranked by expected lives saved per unit of effort before the peak, each was given a named payer and trigger or marked unfunded, and the funding placements were reconciled against the ranked list.
- **Verification.** Separate verifier agents checked every recommendation on three tests: evidence (the primary source and its effect size), transferability (whether that evidence came from somewhere like here), and feasibility (whether it can be done here before the peak). Financial actions were also checked for availability in this jurisdiction, a named payer, a trigger threshold, and whether they already exist. The funding envelope was checked for real recipients, published unit costs, and duplication, then re-summed.
- **Verdicts.** *Confirmed*: evidence, transferability, and feasibility all hold. *Partially confirmed*: holds after the corrections shown. *Unverified*: could not be checked to a primary source. *Contraindicated*: would do harm or waste the window here.
- **Adaptation Inventory.** Could not be consulted: its download sits behind a sign-up form asking for an email address, and the on-page tool does not render without a browser. No action is marked as having or lacking a precedent there.
- **Health wording.** Passages where a wording error could hurt someone (fans and cooling for older people, chlorinating water, check-ins on older residents, and emergency supplies) are marked "* Suggested formulation: check with a health worker before sharing." They were checked against published sources, not by a health professional. They are suggestions, not medical advice, and anyone passing them on should confirm them with a doctor, pharmacist, nurse, or community health worker first. The marker was added on 2 November 2026.

* Suggested formulation: check with a health worker before sharing.

- **Rules.** No cost is derived by dividing a published total, and no local figure is invented; where data is thin, the report says so and shows the containing geography's number with its arithmetic. Every source is listed below with its verification status.
- **Limits.** The panel and verifiers are AI research agents working from web-available sources, not human experts, and no one in this place was interviewed. In this run the Impact Map could not be queried by script, so the impact region is unresolved; several verifiers exhausted the shared web-search budget, which is why some items remain unverified rather than confirmed; and a few official sources (CNE's declarations register, IFRC, SINIRUBE) returned access errors.

References · CNE alert, 18 Sep 2026 — La Nación [confirmed] · Infobae 18 Sep [confirmed] · AyA national plan — Infobae 28 Sep [corrected] · 2026 rain decree 45910-MP — Delfino [confirmed; no drought decree found] · ASADA history — Ballena Tales [confirmed] · Río Uvita line — TV Sur [confirmed] · Rainwater pilot — LatinClima 2019 [corrected] · Population — INEC 2024 estimate via Telencuestas [secondary] · ARESEP ASADA tariffs — ARESEP [confirmed] · Availability letters — CRHoy on 2018-002898 [confirmed] · Well drilling — DE-43053-MINAE [confirmed] · Tank prices — EPA, Mayzap [confirmed] · Rainfall — aggregator climatology [not IMN] · Camino Verde — BMJ 2015 [confirmed] · Dengue lag — Lowe et al. 2021 [confirmed, modelling] · MinSalud vector programme — MinSalud 2025 [confirmed] · Lids — Vietnam trial [confirmed], Central America [title only] · Aedes in tanks — Beebe 2009 [confirmed] · Intermittent supply — Ercumen 2015 [confirmed] · Heat decree — Decreto 39147-S-TSS [confirmed] · Adelante — Hansson 2025, ROI [corrected] · Ahmedabad — Hess 2018 [confirmed, low transferability] · Fans — Meade et al. 2024 [claim contradicted] · Nate — La Nación, FloodList [tolls vary] · César — IPS 1996 · Alma — Wikipedia [secondary] · 2014 decree — ReliefWeb · 2019 decree — Radio Montecarlo [confirmed] · 2023–24 rationing — Semanario Universidad · ONI — NOAA CPC [confirmed] · IMN ENSO — IMN · Reef — Alvarado et al. 2005 [corrected], NOAA Coral Reef Watch [confirmed] · Osa CME — Municipalidad de Osa [exists; members not listed] · SINAC brigades — SINAC [Osa not shown] · IMAS emergencies — IMAS [confirmed] · Cat DDO — World Bank program document [confirmed; drought eligible] · JUDESUR — JUDESUR [covers Osa] · IFRC — GO appeals, flood EAP [confirmed] · CCRIF — membership [confirmed] · Envision — 2026 statement [confirmed; no dollar commitment] · ADI de Uvita — SOMOS [exists; no audited accounts found] · Kremer et al. — NBER w30835 [misquoted; dropped]
Exposure data · Climate Impact Lab, 451,000 at Risk · Impact Map · Adaptation Inventory (not consulted: download gated)
Generated 28 September 2026 for Uvita, Bahía Ballena district, Osa, Costa Rica. El Niño Ready is volunteer-run and is not affiliated with Climate Week NYC, The Climate Group, or Columbia University.

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